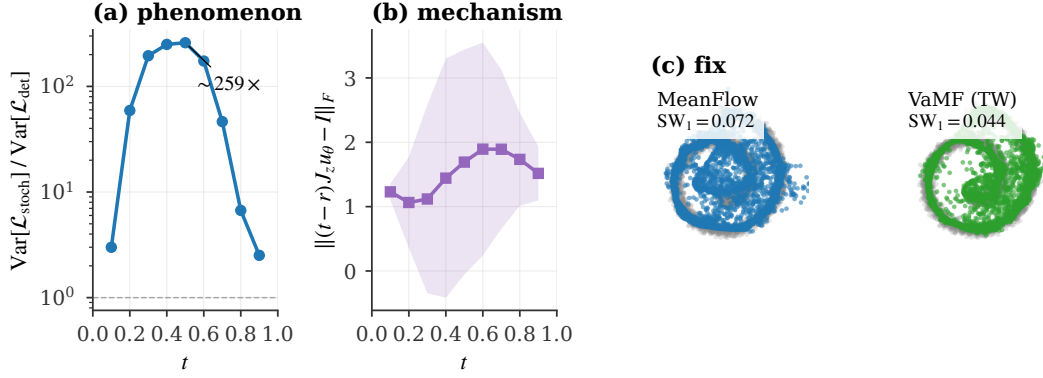


On Variance Reduction in Learning Mean Flows

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Abstract

Mean Flows learn promising distillation-free, one-step generation through optimizing the identity between average and instantaneous velocity field. It is, however, notorious for its non-decreasing, high-variance loss. We investigate the phenomenon and associates these manifestations to a fundamental issue in learning two-parameter average velocity field, that is, the existence of conditional-marginal velocity difference. Based on this discovery, we derive a concrete yet simple method, *Variance-Aware MeanFlow* (VaMF), to reduce the gradient variances and promote both efficiency and performance of the model.

Add experiment results later...

1 Introduction

Flow-based generative models [1, 2, 3, 4, 5, 6, 7] learn continuous-time transformations between a simple prior and a target data distribution. Recent conditional flow matching [5, 6] and stochastic interpolant [7, 8] frameworks enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [9, 10, 11, 12, 13, 14, 15] and variational autoencoders [16]. Like diffusion models, however, they require iterative integration at inference time, motivating methods that reduce the number of function evaluations to one or few steps [17]. Existing approaches include progressive distillation [18], distribution matching [19, 20], consistency models [21, 22], and flow map matching [23], but most require a pre-trained teacher model.

MeanFlow [24] offers an appealing *distillation-free* alternative by learning a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$ through a total-derivative identity. In principle, this identity provides exact self-supervision: the compound prediction $\mathbf{u} + (t-r) \frac{d}{dt} \mathbf{u}$ should match the instantaneous velocity at convergence, requiring no external teacher. In practice, however, MeanFlow training is plagued by non-decreasing loss and prohibitively high gradient variance [25, 26]. Concurrent works diagnose this from different angles: AlphaFlow [25] identifies gradient conflict between loss components; Re-MeanFlow [27] attributes instability to trajectory curvature; and Improved MeanFlow [26] traces it to Jacobian amplification in the JVP. Each proposes a different fix— α -scheduling, trajectory rectification, and a separate velocity head—treating the problems as isolated.

We show that these are *three manifestations of a single quantity*: the material derivative $\frac{D\mathbf{v}}{Dt} := \partial_t \mathbf{v} + (\nabla_{\mathbf{x}} \mathbf{v}) \mathbf{v}$ of the marginal velocity field, which measures how rapidly the velocity changes along its own trajectories. We call this the **curvature-variance principle**. Specifically, we prove that: (i) the curvature gap $\Delta := \mathbf{u} - \mathbf{v}$ satisfies $\Delta \approx -\frac{(t-r)}{2} \frac{D\mathbf{v}}{Dt}$, so it is the local flow acceleration that causes the average and instantaneous velocities to diverge; (ii) the per-step gradient variance is amplified by a Jacobi factor $\text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^\top)$ where $\mathbf{J} = (t-r)\partial_{\mathbf{z}}\mathbf{u}_\theta - \mathbf{I}_d$, and the stop-gradient operator creates a semi-gradient gap that blinds the optimizer to this amplification; and (iii) the acceleration-to-noise ratio $\text{ANR} = (t-r)\|\frac{D\mathbf{v}}{Dt}\|/\sqrt{\text{Tr}(\Sigma_{\mathbf{v}'})}$ governs the optimal sampling distribution over intervals, connecting scheduling heuristics in prior work to a principled criterion.

From this analysis we derive **Variance-Aware MeanFlow (VaMF)**, a training framework that addresses all identified failure modes without architectural modification. VaMF combines: (1) an *EMA velocity anchor* that provides a deterministic JVP tangent, eliminating Jacobian variance amplification; (2) a *flow matching anchor* that supervises the boundary condition, preventing tangent bias without a separate network head; and (3) a *heteroscedastic variance head* that learns per-sample weighting, automatically subsuming inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling. Our contributions are:

- A theoretical framework that unifies gradient conflict, Jacobian amplification, and the curvature bottleneck under the curvature-variance principle, including the first formal quantification of the MeanFlow semi-gradient gap (sections 3.1 and 3.2).
- The VA-MeanFlow training framework that eliminates variance amplification and tangent bias through EMA anchoring, FM anchoring, and a learned variance head—without modifying the backbone (section 3.3).
- A formal comparison between the d/dt and d/ds formulations, revealing the tradeoff between tangent estimation error and architectural compatibility (Appendix D).

2 Preliminary

Flow-Matching. Let $\mathcal{X} \subset \mathbb{R}^d$ be the space of random variables $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{x}$ a data sample as the realized state. The objective of flow-based generative models is to learn a time-dependent diffeomorphic map $\psi(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target data distribution $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$. Specifically, this is achieved by learning a *time-dependent velocity field* $\mathbf{v}(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$:

$$\mathbf{v}(\mathbf{x}, t) \triangleq \frac{d}{dt}\psi(\mathbf{x}, t), \quad \psi(\mathbf{x}, 0) = \mathbf{x}_0. \quad (1)$$

The map $\psi(\mathbf{x}, t)$ induces a *time-dependent probability density path* $p(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}_{>0}$, with $\int_{\mathcal{X}} p(\mathbf{x}, t) d\mathbf{x}, \forall t \in [0, 1]$, through the pushforward measure $p(\mathbf{x}, t) \triangleq [\psi]_{\#}(p(\mathbf{x}, 1))$, where the pushforward operator $[\cdot]_{\#}$ is defined by the change-of-variable operation

$$[\psi]_{\#}(p(\mathbf{x}, 1)) = p(\psi^{-1}(\mathbf{x}, t)) \left| \det \left[\frac{\partial}{\partial \mathbf{x}} \psi^{-1}(\mathbf{x}, t) \right] \right|. \quad (2)$$

The sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation*, which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t} p(\mathbf{x}, t) + \text{div}(p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0, \quad (3)$$

where $\text{div}(f)$ denotes the divergence of function f . However, the exact marginal velocity field $\mathbf{v}(\mathbf{x}, t)$ is generally *inaccessible*, especially in a high-dimensional sample space. The optimal-transport conditional flow-matching (OT-CFM) [5, 6] alternatively learns a conditional velocity field $\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0; \theta) \triangleq \mathbf{v}((1-t)\mathbf{x}_0 + t\mathbf{x}_1; \theta)$ by minimizing a loss function:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), \mathbf{x}_0 \sim p_{\text{data}}, \mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)} \|\mathbf{v}(\mathbf{x}, t; \theta) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2. \quad (4)$$

Herein, $t \sim \mathcal{U}(0, 1)$ is a scalar sample from the uniform distribution defined on the range of $[0, 1]$ and $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$ is a white noise sample from the standard normal distribution defined in $\mathbb{R}^d$. Meanwhile, there exists a relationship between the marginal and conditional velocity field

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0, t | \mathbf{x})} [\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)], \quad (5)$$

which is the key to the validity of surrogate loss in equation 4. See Appendix B for its derivation.

One-step Mean Flows. The issue with diffusion [11, 13] and flow-matching models [5] is the integration. Evaluating the velocity field multiple times is not only expensive but can also lead to numerical instability. A Mean Flow model [24] instead learns a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$ to enable efficient sampling. It leverages the identity between the average and marginal velocity field given by

$$(t - r)\mathbf{u}(\mathbf{x}, r, t) = \int_{\tau=r}^t \mathbf{v}(\mathbf{x}, \tau) d\tau, \quad (6)$$

where r and t are scalars, the start and end time steps along the probability density path. Taking the total derivative with respect to the terminal time step t yields

$$\mathbf{u}(\mathbf{x}, r, t) + (t - r) \frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \mathbf{v}(\mathbf{x}, t), \quad (7)$$

where the total derivative of the average velocity field with respect to t can be expressed as a Jacobi-vector Product (JVP)

$$\frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_r \mathbf{u} \cdot \frac{dr}{dt} + \partial_t \mathbf{u} \cdot \frac{dt}{dt} = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_t \mathbf{u} \quad (8)$$

We abuse the notation and express the right-hand side by $\text{JVP}(\mathbf{u}, (\mathbf{x}, r, t), (\mathbf{v}(\mathbf{x}, t), 0, 1))$. In the original MeanFlow [24], the authors replace both the marginal velocity field in the right-hand side of equation 7 and inside the JVP with the conditional velocity field, and derive the MeanFlow loss function given by

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\left\| \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg} [\text{JVP}(\mathbf{u}_{\boldsymbol{\theta}}, (\mathbf{z}, r, t), (\mathbf{v}_{\text{cond}}, 0, 1))] - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (9)$$

where $r \sim \mathcal{U}(0, 1)$, $t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$, $\mathbf{z} = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$, and $\text{sg}[\cdot]$ is the stop-gradient operator. The authors highlight that stopping gradients in JVP helps eliminate the need for “double backpropagation” and high-order derivatives.

3 Variance-aware Mean Flows

It is well known to the community that the MeanFlow loss in equation 9 is *non-decreasing* and has *high-variance gradients* [25, 26]. In the following sections, we attribute the issue to replacing the marginal velocity field with the conditional field, which introduces a bias that invalidates the loss function. Our research questions are as follows:

RQ1: Source of bias and variance in MeanFlow training.

Is it valid to use a conditional velocity field in place of a marginal field in the MeanFlow loss?

RQ2: Variance reduction in learning MeanFlow.

What is the variance structure of the MeanFlow gradients? How can we then reduce the variances in training MeanFlow models?

We approach these questions from a theoretical perspective in Section 3.1 and quantify the empirical variances by *flow acceleration* in Section 3.2. Through the analysis, we reveal a fundamental bias as the root cause of the original non-decreasing MeanFlow loss. To mitigate the issue, we propose the Variance-Aware MeanFlow (VaMF) in Section 3.3, a training framework that reduces gradient variance without modifying the backbone architecture. See Appendix C for full proofs of all our theorems in this section.

3.1 Bias and Variances in Learning Mean Flows

In this section, we establish a theoretical analysis of the consequences of replacing the inaccessible marginal velocity $\mathbf{v}(\mathbf{x}, t)$ by the conditional field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$ in both the JVP tangent and the regression target. We can apply the Reynolds decomposition of the field and let $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) =$

$\mathbf{v}(\mathbf{x}, t) + \mathbf{v}'$, where $\mathbf{v}'$ is a zero-mean fluctuation from the marginal velocity induced by optimal transport between latent $\mathbf{x}_1$ and data $\mathbf{x}_0$. The regression loss equation 9 expands to

$$\begin{aligned}\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) &= \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_{\boldsymbol{\theta}} + (t-r) \text{sg} [\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v}_{\text{cond}} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}_{\text{cond}} \right\|_2^2 \right] \\ &= \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_{\boldsymbol{\theta}} + (t-r) \text{sg} [\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}(\mathbf{x}_t, t) + \text{sg}[\mathbf{J}]\mathbf{v}' \right\|_2^2 \right] \\ &= \mathbb{E}_{r,t,\mathbf{x}_t,\mathbf{x}_1} \left[\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\left\| \underbrace{\mathbf{u}_{\boldsymbol{\theta}} + (t-r) \text{sg} [\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}(\mathbf{x}_t, t)}_{\text{mean-field residual } \mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}(\mathbf{x}_t, t)} + \text{sg}[\mathbf{J}]\mathbf{v}' \right\|_2^2 \right] \right].\end{aligned}\quad (10)$$

Herein, we again abuse the notations and let $\mathbf{J} \triangleq (t-r)\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$ denote a Jacobi factor of $\mathbf{u}_{\boldsymbol{\theta}}$ given sample $\mathbf{z} = \mathbf{x}_1 - \mathbf{x}_0$. In this decomposition, we notice that the original MeanFlow regression loss can be expressed as the sum of a deterministic mean-field residual $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}(\mathbf{x}_t, t)$ and a sample-based Jacobi-velocity-product $\mathbf{J}\mathbf{v}'$. If we further expand the inner expectation, based on equation 5, we have $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}'] = 0$ and the cross term vanishes, resulting in

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathcal{L}_{\text{MF}}(\boldsymbol{\theta})] = \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\left\| \mathbf{r}(\mathbf{x}_t, t) + \text{sg}[\mathbf{J}\mathbf{v}'] \right\|_2^2 \right] = \|\mathbf{r}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|_2^2 + \text{sg}[\text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^{\top})], \quad (11)$$

where $\Sigma_{\mathbf{v}'} = \text{Cov}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}']$ is the conditional velocity covariance and $\mathcal{L}_{\text{MF}}(\boldsymbol{\theta})$ is the per-step loss. We can prove that the magnitude of the average-velocity Jacobian amplifies the per-step variance of the gradient.

Theorem 1 (Jacobian Variance Amplification.). *The Jacobi factor amplifies the variances of the original MeanFlow loss*

$$\boxed{\text{Var} [\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) \mid \mathbf{x}_t] \propto \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^{\top})}$$

Theorem 1 explains the instability and high variance observed in training the MeanFlow model. With large intervals $(t-r)$ or large norm of the Jacobi term $\|\partial_{\mathbf{x}} \mathbf{u}_{\boldsymbol{\theta}}\|_2^2$, the per-step gradient is dominated by the noise term $\mathbf{J}\mathbf{v}'$, downweighting the mean-field signal from $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}(\mathbf{x}_t, t)$.

Crucially, the $\mathbf{J}\mathbf{v}'$ amplification arises because $\mathbf{v}'$ enters through the JVP tangent, where the spatial Jacobian $(t-r)\partial_{\mathbf{z}} \mathbf{u}_{\boldsymbol{\theta}}$ multiplies it. If the tangent were replaced by *any* deterministic estimate $\hat{\mathbf{v}}$, the fluctuation $\mathbf{v}'$ would only appear in the regression target $\mathbf{v}_{\text{cond}} = \mathbf{v} + \mathbf{v}'$, entering the residual linearly rather than through $\mathbf{J}$. In that case, the per-step loss reduces to $\|\tilde{\mathbf{r}}_{\boldsymbol{\theta}}\|_2^2 + \text{Tr}(\Sigma_{\mathbf{v}'})$, where $\tilde{\mathbf{r}}_{\boldsymbol{\theta}}$ is a deterministic residual—the variance drops to the irreducible level $\text{Tr}(\Sigma_{\mathbf{v}'})$ without any Jacobian amplification.

Furthermore, since the inner expectation includes the Jacobian amplified trace, we find that the stop-gradient operator creates a *semi-gradient gap*.

Theorem 2 (Semi-Gradient Gap). *The gradients of MeanFlow loss with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [\nabla_{\boldsymbol{\theta}} (\partial_{\mathbf{x}} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_{\boldsymbol{\theta}}) \cdot \mathbf{r}_{\boldsymbol{\theta}}]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}}$$

Herein, the stop-gradient operator eliminates the trace term from the expected gradient and $\mathbf{r}_{\boldsymbol{\theta}} \rightarrow \mathbf{0}$ at convergence. Therefore, the variance-driven term dominates the semi-gradient gap near convergence. $\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^{\top})$ forces the system to minimize $\|(t-r)\partial_{\mathbf{x}} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d\|$, essentially pushes $\partial_{\mathbf{x}} \mathbf{u}_{\boldsymbol{\theta}} \rightarrow \frac{1}{t-r}\mathbf{I}_d$ to minimize the magnitude of $\mathbf{J}$. On the contrary, stop-gradient operator blinds variance-driven gradient, and as a result, $\text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}\mathbf{J}^{\top})$ can potentially increase faster than the magnitude of decrease in $\|\mathbf{r}_{\boldsymbol{\theta}}\|_2^2$, leading to the non-decreasing loss.

Both failure modes—Jacobian variance amplification (theorem 1) and the unobserved variance growth from the semi-gradient gap (theorem 2)—share a common root. Define the *curvature gap*

$$\Delta(\mathbf{x}_t, r, t) := \mathbf{u}(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t), \quad (12)$$

the difference between the average and instantaneous velocity fields. Rearranging the MeanFlow identity equation 7 gives $(t-r)\frac{d}{dt}\mathbf{u} = -\Delta$. The mean-field residual in equation 10 can therefore be rewritten as $\mathbf{r}_\theta^{\text{sg}} = \Delta_\theta + (t-r)\frac{d}{dt}\mathbf{u}_\theta$, where $\Delta_\theta := \mathbf{u}_\theta - \mathbf{v}$. At convergence the total derivative must exactly cancel the curvature gap ($\Delta_\theta + (-\Delta_\theta) = 0$), so the training signal $\mathbf{r}_\theta^{\text{sg}}$ is a small difference of two quantities whose magnitudes scale with $\|\Delta\|$. When $\|\Delta\|$ is large, this cancellation is delicate: the gradient signal from $\mathbf{r}_\theta$ is weak relative to the noise $\mathbf{J}\mathbf{v}'$ from theorem 1, and the semi-gradient gap from theorem 2 prevents the optimizer from controlling the resulting variance growth. This raises a natural question: how large is Δ in practice, and what governs its magnitude?

3.2 Flow Acceleration

The curvature gap $\Delta(\mathbf{x}_t, r, t)$ defined in equation 12 controls the severity of both Jacobian variance amplification and the semi-gradient gap. We now quantify Δ using flow acceleration, revealing its dependence on the material derivative of the velocity field.

Theorem 3 (Flow Acceleration Expansion of the Expectation Gap). *Let $\mathbf{v}(\mathbf{x}, t) \in C^2$ be twice continuously differentiable. The expectation gap admits a Taylor series at $\mathbf{x}_t$:*

$$\Delta(\mathbf{x}_t, r, t) = -\frac{(t-r)}{2} \frac{D\mathbf{v}}{Dt}(\mathbf{x}_t, t) + \frac{(t-r)^2}{6} \frac{D^2\mathbf{v}}{Dt^2}(\mathbf{x}_t, t) + \mathcal{R}_3(\mathbf{x}_t, r, t), \quad (13)$$

where $\frac{D\mathbf{v}}{Dt}(\mathbf{x}, t) \triangleq \partial_t \mathbf{v}(\mathbf{x}, t) + (\nabla_{\mathbf{x}} \mathbf{v}(\mathbf{x}, t)) \cdot \mathbf{v}(\mathbf{x}, t)$ is the material dervative (a.k.a. flow acceleration) and $\mathcal{R}_3(\mathbf{x}, r, t)$ is the third-order remainder of the Taylor series.

Herein, it is reasonable to assume $\mathbf{v}(\mathbf{x}, t) \in C^2$ since existing work primarily uses a U-Net [28, 12] or transformer backbone with twice continuously differentiable activation functions (e.g., the *GELU* activation [29]) to learn the velocity field. Theorem 3 establishes a theoretical perspective that bridges the points discussed in the existing works. Specifically, methods with rectified flows [30] straighten the marginal ODE trajectory, thereby enforcing near-zero flow accelerations. By theorem 3, this pushes $\Delta(\mathbf{x}, r, t) \rightarrow 0$ and eliminates the root cause for training instability.

Meanwhile, methods that employ progressive scheduling [26] address the problem by initially training with a small interval $r \rightarrow t$. According to theorem 3, the expectation gap grows at a rate factor $\frac{(t-r)}{2} \left\| \frac{D}{Dt}(\mathbf{x}, r, t) \right\|$ and hence it is reasonable to constrain the maximum interval during training by bounding this factor. The expansion enables us to establish a Lipschitz bound on the expectation gap.

In addition, the relationship between the expectation gap and the flow acceleration reveals a challenge in the intermediate time step t , where the flow acceleration maximizes due to the intersection of conditional trajectories. The issue is exacerbated for high-resolution, multi-modal data, where marginal velocity field changes occur rapidly. See Appendix C for a complementary Lipschitz bound on Δ .

Together with the variance analysis in section 3.1, the material derivative expansion yields three design principles for stabilizing MeanFlow training:

- D1 Eliminate variance amplification.** As shown above, $\mathbf{v}'$ enters through the JVP tangent and is amplified by $\mathbf{J}$ (theorem 1). Replacing the stochastic tangent with a deterministic estimate moves $\mathbf{v}'$ outside the JVP, reducing variance to the irreducible level $\text{Tr}(\Sigma_{\mathbf{v}'})$.
- D2 Correct tangent bias.** Any deterministic tangent that differs from $\mathbf{v}(\mathbf{x}_t, t)$ introduces a compound-prediction bias. The semi-gradient gap (theorem 2) blinds the optimizer to this bias, so auxiliary boundary supervision is needed.
- D3 Adapt to local curvature.** Since $\|\Delta\| \propto (t-r) \left\| \frac{D\mathbf{v}}{Dt} \right\|$ by theorem 3, per-sample difficulty varies with both interval length and flow acceleration. Optimal weighting should downweight high-curvature, long-interval samples where the training signal is weakest.

3.3 Variance-Aware MeanFlow

We now instantiate these three design principles into a concrete training framework—Variance-Aware MeanFlow (VaMF)—that addresses all identified failure modes without modifying the backbone architecture.

3.3.1 Velocity Anchoring (D1)

To eliminate the Jacobian variance amplification identified in theorem 1, we replace the stochastic JVP tangent $\mathbf{v}_{\text{cond}}$ with a deterministic estimate. Following IMF [26], we use a Polyak averaging [31] (*a.k.a. exponential moving average (EMA)*) [32] of the model parameters as the tangent. Specifically, the new loss writes

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) = \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_\theta + (t-r) \text{sg} [\text{JVP}(\mathbf{u}_\theta, (\mathbf{z}, r, t), (\mathbf{u}_{\bar{\theta},t}, 0, 1))] - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (14)$$

where $\mathbf{u}_{\bar{\theta},t} = \mathbf{u}(\mathbf{x}_t, t, t)$ and $\bar{\theta}$ is updated by $\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ with $\mu \in [0, 1]$.

Remark 4. The Polyak averaging slowly evolves $\mathbf{u}_{\bar{\theta}}$ to the online model $\mathbf{u}_\theta$. In the early stage of the training, both $\mathbf{u}_\theta$ and $\mathbf{u}_{\bar{\theta}}$ are poor estimators. Still, the anchoring provides a direct supervision that rapidly improves $\mathbf{u}_\theta$, and hence indirectly improves $\mathbf{u}_{\bar{\theta}}$.

3.3.2 The VA-MeanFlow Objective (D2)

The EMA tangent solves D1 but introduces a transient bias when $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) \neq \mathbf{v}(\mathbf{x}_t, t)$, leaving D2 unaddressed. To correct this, we add a *flow matching anchor* that provides direct velocity supervision near the boundary $r \rightarrow t$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1} \left[\left\| \mathbf{u}_\theta(\mathbf{x}_t, t-\delta, t) - \mathbf{v}_{\text{cond}} \right\|_2^2 \right], \quad (15)$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ , the average velocity over $[t-\delta, t]$ approximates the instantaneous velocity, so equation 15 is asymptotically equivalent to equation 4 without a separate network head. The full **VA-MeanFlow** objective is:

$$\mathcal{L}_{\text{VA-MF}}(\theta) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) + \lambda \mathcal{L}_{\text{FM}}(\theta), \quad (16)$$

where $\lambda > 0$ balances the two terms. The following propositions characterize the gradient structure.

Proposition 5 (Gradient of VA-MF Loss). *The gradient of $\mathcal{L}_{\text{MF}}^{\text{EMA}}$ takes the form $\nabla_\theta \mathcal{L}_{\text{MF}}^{\text{EMA}} = 2 \mathbb{E}[\nabla_\theta \mathbf{u}_\theta(\mathbf{x}_t, r, t) \cdot \tilde{\mathbf{r}}^{\text{EMA}}]$, where $\tilde{\mathbf{r}}^{\text{EMA}} := \mathbf{V}_\theta^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is the mean-field residual under the EMA tangent. The conditional velocity variance does not contribute.*

Proposition 6 (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as $\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_\theta + (t-r) \partial_{\mathbf{z}} \mathbf{u}_\theta(\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t))$, where $\mathbf{r}_\theta$ is the true MeanFlow residual from equation 10. At convergence, $\mathbf{r}_\theta \rightarrow \mathbf{0}$ and the FM anchor drives $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.*

Proposition 5 confirms that the VA-MF gradient has a first-order structure with no Jacobian-mediated variance amplification (cf. theorem 1). Proposition 6 shows the transient bias is bounded by $(t-r) \|\partial_{\mathbf{z}} \mathbf{u}_\theta\| \|\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)\|$ and shrinks as the EMA velocity improves.

3.3.3 Heteroscedastic Extension (D3)

With D1 and D2 resolved, the remaining per-sample variance stems from the conditional velocity fluctuation $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, whose magnitude $\text{Tr}(\Sigma_{\mathbf{v}'})$ varies with both t and $\mathbf{x}_t$. By theorem 3, the curvature gap contributes an additional $(t-r)^2 \left\| \frac{\text{D}\mathbf{v}}{\text{D}t} \right\|^2 / 4$ to the per-sample difficulty, so the total difficulty depends on both the irreducible noise $\text{Tr}(\Sigma_{\mathbf{v}'})$ and the curvature-driven term—precisely the structure that D3 calls for. Moreover, this difficulty is not spatially uniform: different regions of a natural image (flat backgrounds, sharp edges, fine textures) have markedly different local predictability. We capture this local structure by extending the model with a *per-pixel-per-channel* variance output $\sigma_\theta^2 : \mathbb{R}^d \times [0, 1]^2 \rightarrow \mathbb{R}_{>0}^d$ and replacing the MSE with a heteroscedastic diagonal-Gaussian negative log-likelihood [33, 34]:

$$\mathcal{L}_{\text{MF}}^{\text{NLL}}(\theta) = \mathbb{E}_{t,r,\mathbf{x}_0,\mathbf{x}_1} \left[\frac{1}{2} \sum_{i=1}^d \left(\frac{(V_{\theta,i}^{\text{EMA}} - \mathbf{v}_{\text{cond},i})^2}{\sigma_{\theta,i}^2(\mathbf{x}_t, r, t)} + \log \sigma_{\theta,i}^2(\mathbf{x}_t, r, t) \right) \right], \quad (17)$$

equivalently $\frac{1}{2} (\tilde{\mathbf{r}}^{\text{EMA}\top} \Sigma_\theta^{-1} \tilde{\mathbf{r}}^{\text{EMA}} + \log \det \Sigma_\theta)$ with $\Sigma_\theta = \text{diag}(\sigma_{\theta,1}^2, \dots, \sigma_{\theta,d}^2)$. Here $\sigma_{\theta,i}^2 = \text{softplus}(s_{\theta,i}) + \sigma_{\min}^2$, where s_θ is a *per-location* linear projection from the shared backbone (one log-variance per pixel per channel) and $\sigma_{\min}^2 > 0$ prevents collapse. The full heteroscedastic objective is $\mathcal{L}_{\text{VA-MF}}^{\text{NLL}} = \mathcal{L}_{\text{MF}}^{\text{NLL}} + \lambda \mathcal{L}_{\text{FM}}^{\text{NLL}}$, where $\mathcal{L}_{\text{FM}}^{\text{NLL}}$ applies the same per-pixel Gaussian NLL to the FM anchor prediction using the variance head evaluated at the anchor interval (rather than the plain MSE in equation 15), so that the two terms live on a common per-element NLL scale and λ has a consistent meaning across training.

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$, variance floor $\sigma_{\min}^2$, warm-up end N_w , NLL ramp length N_{ramp}

Initialize $\bar{\theta} \leftarrow \theta$

for $k = 1, 2, \dots$ **do**

Sample $(x_0, x_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$

$x_t \leftarrow (1 - t)x_0 + tx_1$

$(\mathbf{u}_\theta, s_\theta) \leftarrow \text{Network}(x_t, r, t)$; $\sigma_\theta^2 \leftarrow \text{softplus}(s_\theta) + \sigma_{\min}^2 \in \mathbb{R}_{\geq 0}^d$ ▷ per-pixel variance

$\mathbf{v}_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(x_t, t, t)]$ ▷ EMA anchor

$\mathbf{V}_\theta \leftarrow \mathbf{u}_\theta + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_\theta, (x_t, r, t), (\mathbf{v}_{\text{tang}}, 0, 1))]$

$\ell_{\text{MF}}^{\text{MSE}} \leftarrow \|\mathbf{V}_\theta - \mathbf{v}_{\text{cond}}\|_2^2$

$\ell_{\text{MF}}^{\text{NLL}} \leftarrow \frac{1}{2} \sum_{i=1}^d [(V_{\theta,i} - v_{\text{cond},i})^2 / \sigma_{\theta,i}^2 + \log \sigma_{\theta,i}^2]$

$\alpha_k \leftarrow \text{clip}((k - (N_w - N_{\text{ramp}})) / N_{\text{ramp}}, 0, 1)$ ▷ MSE $\rightarrow$ NLL ramp

$\ell_{\text{MF}} \leftarrow (1 - \alpha_k) \ell_{\text{MF}}^{\text{MSE}} + \alpha_k \ell_{\text{MF}}^{\text{NLL}}$

$(\mathbf{u}_\theta^\delta, \sigma_\theta^{2,\delta}) \leftarrow \text{Network}(x_t, t - \delta, t)$ ▷ FM anchor forward

$\ell_{\text{FM}} \leftarrow \frac{1}{2} \sum_{i=1}^d [(\mathbf{u}_{\theta,i}^\delta - v_{\text{cond},i})^2 / \sigma_{\theta,i}^{2,\delta} + \log \sigma_{\theta,i}^{2,\delta}]$ ▷ NLL-consistent FM anchor

$\theta \leftarrow \theta - \eta \nabla_\theta (\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$

$\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu) \theta$ ▷ EMA update

end for

return θ ▷ Generate: $\hat{x}_0 = x_1 - \mathbf{u}_\theta(x_1, 0, 1)$

Proposition 7 (Variance Head as Automatic Per-Element Weighting). *Under the heteroscedastic loss equation 17: (a) the mean-parameter gradient is the per-element-weighted form of proposition 5 with $1/(2\sigma_{\theta,i}^2)$ as the learned weight on the i -th pixel-channel residual; (b) for fixed mean parameters, the optimal per-element variance is $\sigma_i^{2*} = \mathbb{E}[(\tilde{\mathbf{r}}_i^{\text{EMA}})^2] + \text{Var}(\mathbf{v}'_i)$, reducing to $\text{Var}(\mathbf{v}'_i)$ at convergence of the mean head; (c) each $\sigma_{\theta,i}^2$ grows with $(t - r)$ via the pixel-local curvature gap (theorem 3), automatically downweighting hard intervals and hard spatial regions without an explicit schedule.*

The per-pixel variance head replaces inverse-variance weighting, progressive scheduling, and EMA-based variance proxies with a single learned spatially-varying map, and simultaneously captures the spatial heteroscedasticity of natural images that a scalar variance cannot represent. At generation time, only $\mathbf{u}_\theta(x_1, 0, 1)$ is needed. The complete training procedure is given in Algorithm 1; see Appendix D for the connection to the acceleration-to-noise ratio and an MSE variant without the variance head.

Remark 8 (MSE-to-NLL ramp). Switching abruptly from MSE to NLL at step N_w leaves the variance head with zero gradient throughout the warm-up, so every $\sigma_{\theta,i}^2$ must traverse from its initialization to its optimum at a single step. We instead linearly interpolate $\ell_{\text{MF}} \leftarrow (1 - \alpha_k) \ell_{\text{MF}}^{\text{MSE}} + \alpha_k \ell_{\text{MF}}^{\text{NLL}}$ with α_k ramping from 0 to 1 over $[N_w - N_{\text{ramp}}, N_w]$, giving the variance head non-zero gradient during the transition and avoiding a discontinuity in the loss landscape.

4 Related Work

Flow Matching. Continuous normalizing flows (CNFs) [1, 4] parameterize a time-dependent velocity field and integrate it to transform a base distribution, extending earlier normalizing-flow approaches [2, 3]. Conditional flow matching [5, 6] and stochastic interpolants [7, 8] enable simulation-free training by exploiting the identity $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)]$, replacing the intractable marginal velocity with conditional samples. Minibatch optimal transport [6, 35] further straightens conditional paths. Scaling rectified flow transformers to high resolution [36] demonstrates the practical power of this paradigm. Our analysis extends the framework to the two-parameter average velocity setting, where the conditional expectation identity breaks down due to trajectory curvature (theorem 3).

One-Step Generation. Reducing the number of function evaluations to one is a central goal. DDIM [17] first enabled deterministic sampling with larger step sizes from diffusion models. Consistency models [21, 37, 22] enforce that all points on a PF-ODE trajectory map to the same endpoint, with extensions to trajectory consistency [38] and easy fine-tuning [39]. Distillation-based methods [18, 19, 20, 40] achieve strong FIDs [41] on standard benchmarks [42] but require a pre-trained teacher. InstaFlow [43] combines rectified flow with distillation for text-to-image one-step generation. Flow map matching [23] and shortcut models [44] provide principled frameworks for two-time maps. Recent work unifies consistency and flow matching [45, 46], and FACM [47] anchors consistency training with a flow matching loss—a mechanism related to our FM anchor, though within a different training framework.

MeanFlow and Variants. MeanFlow [24] learns the average velocity field via a total-derivative identity, enabling distillation-free one-step generation. AlphaFlow [25] identifies gradient conflict between the flow matching and total consistency components, proposing α -scheduling to manage the tradeoff. Improved MeanFlow [26] traces instability to the stochastic JVP tangent and introduces a separate velocity head with stop-gradient. Re-MeanFlow [27] addresses trajectory curvature through a reflow pre-processing step. Terminal Velocity Matching (TVM) [48] sidesteps the spatial Jacobian entirely by differentiating w.r.t. the terminal time, and Functional Mean Flows [49] independently establishes the marginal-conditional gap for average velocities in a Hilbert space setting. Physics-informed distillation [50] derives a related PDE-based loss for learning flow maps. Our curvature-variance principle unifies these diagnostics—our method addresses all failure modes without architectural changes, separate heads, or pre-training.

Variance Reduction in Flow Training. Several concurrent works address variance in flow-based training. Stable Velocity [51] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [52] reduces variance through temporal pair coupling; and preconditioned flow matching [53] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [54] argue that stochastic target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification is specific to the MeanFlow JVP, not the flow matching loss itself. Rectified flows [30, 55] reduce trajectory curvature by iterative straightening, which by theorem 3 drives $\frac{Dv}{Dt} \rightarrow 0$.

5 Experiments

6 Conclusion

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetur eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor. Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

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References

- [1] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

- [2] Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1530–1538, 2015.
- [3] Laurent Dinh, Jascha Sohl-Dickstein, and Samy Bengio. Density estimation using Real-NVP. In *International Conference on Learning Representations*, 2017.
- [4] Will Grathwohl, Ricky T. Q. Chen, Jesse Bettencourt, Ilya Sutskever, and David Duvenaud. FFJORD: Free-form continuous dynamics for scalable reversible generative models. In *International Conference on Learning Representations*, 2019.
- [5] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023.
- [6] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguët, Yanlei Zhang, Jarrod Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport, 2024.
- [7] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants, 2023.
- [8] Michael Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.
- [9] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2256–2265, 2015.
- [10] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [11] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020.
- [12] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171, 2021.
- [13] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.
- [14] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [15] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [16] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014.
- [17] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.
- [18] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.
- [19] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Frédo Durand, William T. Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6613–6623, June 2024.
- [20] Tianwei Yin, Michaël Gharbi, Taesung Park, Richard Zhang, Eli Shechtman, Frédo Durand, and William T. Freeman. Improved distribution matching distillation for fast image synthesis. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 47455–47487. Curran Associates, Inc., 2024.
- [21] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023.

- [22] Cheng Lu and Yang Song. Simplifying, stabilizing and scaling continuous-time consistency models, 2025.
- [23] Nicholas M. Boffi, Michael S. Albergo, and Eric Vanden-Eijnden. Flow map matching with stochastic interpolants: A mathematical framework for consistency models, 2025.
- [24] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling, 2025.
- [25] Huijie Zhang, Aliaksandr Siarohin, Willi Menapace, Michael Vasilkovsky, Sergey Tulyakov, Qing Qu, and Ivan Skorokhodov. Alphaflow: Understanding and improving meanflow models, 2025.
- [26] Zhengyang Geng, Yiyang Lu, Zongze Wu, Eli Shechtman, J. Zico Kolter, and Kaiming He. Improved mean flows: On the challenges of fastforward generative models, 2025.
- [27] Xinxi Zhang, Shiwei Tan, Quang Nguyen, Quan Dao, Ligong Han, Xiaoxiao He, Tunyu Zhang, Chengzhi Mao, Dimitris Metaxas, and Vladimir Pavlovic. Overcoming the curvature bottleneck in meanflow, 2026.
- [28] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241, 2015.
- [29] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.
- [30] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow, 2022.
- [31] Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. volume 30, pages 838–855, 1992.
- [32] Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised learning results. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [33] David A. Nix and Andreas S. Weigend. Estimating the mean and variance of the target probability distribution. In *Proceedings of the IEEE International Conference on Neural Networks*, pages 55–60, 1994.
- [34] Alex Kendall and Yarin Gal. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [35] Gabriel Peyré and Marco Cuturi. Computational optimal transport: With applications to data science. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [36] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Naveen Rafi, Tim Shafir, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024.
- [37] Yang Song and Prafulla Dhariwal. Improved techniques for training consistency models, 2023.
- [38] Dongjun Kim, Chieh-Hsin Lai, Wei-Hsiang Liao, Naoki Murata, Yuhta Takida, Toshimitsu Uesaka, Yutong He, Yuki Mitsufuji, and Stefano Ermon. Consistency trajectory models: Learning probability flow ode trajectory of diffusion, 2024.
- [39] Zhengyang Geng, Ashwini Pople, William Luo, Justin Lin, and J. Zico Kolter. Consistency models made easy, 2024.
- [40] Mingyuan Zhou, Huangjie Zheng, Zhendong Wang, Mingzhang Yin, and Hai Huang. Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 62307–62331. PMLR, 21–27 Jul 2024.
- [41] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [42] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

- [43] Xingchao Liu, Xiwen Zhang, Jianzhu Ma, Jian Peng, and Qiang Liu. InstaFlow: One step is enough for high-quality diffusion-based text-to-image generation, 2024.
- [44] Kevin Frans, Danijar Hafner, Sergey Levine, and Pieter Abbeel. One step diffusion via shortcut models, 2025.
- [45] Ling Yang, Zixiang Zhang, Zhilong Zhang, Xingchao Liu, Minkai Xu, Wentao Zhang, Chenlin Meng, Stefano Ermon, and Bin Cui. Consistency flow matching: Defining straight flows with velocity consistency, 2024.
- [46] Zhangkai Wu, Xuhui Fan, Hongyu Wu, and Longbing Cao. Scot: Unifying consistency models and rectified flows via straight-consistent trajectories, 2025.
- [47] Yansong Peng, Kai Zhu, Yu Liu, Pingyu Wu, Hebei Li, Xiaoyan Sun, and Feng Wu. Facm: Flow-anchored consistency models, 2025.
- [48] Linqi Zhou, Mathias Parger, Ayaan Haque, and Jiaming Song. Terminal velocity matching, 2026.
- [49] Zhiqi Li, Yuchen Sun, Greg Turk, and Bo Zhu. Functional mean flow in hilbert space, 2025.
- [50] Joshua Tian Jin Tee, Kang Zhang, Hee Suk Yoon, Dhananjaya Nagaraja Gowda, Chanwoo Kim, and Chang D. Yoo. Physics informed distillation for diffusion models, 2024.
- [51] Donglin Yang, Yongxing Zhang, Xin Yu, Liang Hou, Xin Tao, Pengfei Wan, Xiaojuan Qi, and Renjie Liao. Stable velocity: A variance perspective on flow matching, 2026.
- [52] Chika Maduabuchi and Jindong Wang. Temporal pair consistency for variance-reduced flow matching, 2026.
- [53] Shadab Ahamed, Eshed Gal, Simon Ghyselincks, Md Shahriar Rahim Siddiqui, Moshe Eliasof, and Eldad Haber. Preconditioned score and flow matching, 2026.
- [54] Quentin Bertrand, Anne Gagneux, Mathurin Massias, and Rémi Emonet. On the closed-form of flow matching: Generalization does not arise from target stochasticity, 2025.
- [55] Sangyun Lee, Zinan Lin, and Giulia Fanti. Improving the training of rectified flows. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 63082–63109. Curran Associates, Inc., 2024.

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A Notations

Table 1 lists all the notations we use in this work.

Table 1: Table of Notations.

Notation	Description
$\mathbf{x}, \mathbf{x}_t$	Random variable / random variable at time t
$\mathbf{x}, \mathbf{x}_t$	Realized state / state at time t : $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
$\mathbf{v}(\mathbf{x}, t)$	Marginal velocity field
$\mathbf{v}_{\text{cond}}$	Conditional velocity: $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}'$	Velocity fluctuation: $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, $\mathbb{E}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$
$\mathbf{u}_{\theta}(\mathbf{x}, r, t)$	Learned average velocity field (two-parameter)
$\mathbf{u}_{\bar{\theta}}$	EMA average velocity with parameters θ
V_{θ}^{EMA}	Compound prediction equation 14
$\mathbf{J}$	Jacobi factor: $\mathbf{J} = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta} - \mathbf{I}_d$
$\Sigma_{\mathbf{v}'}$	Conditional velocity covariance: $\text{Cov}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}']$
σ_{θ}^2	Learned scalar variance output
$\frac{D\mathbf{v}}{Dt}$	Material derivative (flow acceleration): $\partial_t\mathbf{v} + (\nabla_{\mathbf{x}}\mathbf{v})\mathbf{v}$
$\Delta(\mathbf{x}_t, r, t)$	Curvature gap: $\mathbf{u}(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t)$
$\text{sg}[\cdot]$	Stop-gradient operator

B Derivation of Conditional-Marginal Velocity Relationship

We derive equation 5: $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0|\mathbf{x}_t=\mathbf{x})}[\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)]$.

Proof. The marginal velocity field generates the marginal density $p(\mathbf{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \int p(\mathbf{x}, t | \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0, \quad (18)$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\mathbf{x}, t)$. Dividing by $p(\mathbf{x}, t) = \int p(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0$ and applying Bayes' rule $p(\mathbf{x}_0 | \mathbf{x}_t = \mathbf{x}) = p(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0) / p(\mathbf{x}, t)$:

$$\mathbf{v}(\mathbf{x}, t) = \int \mathbf{v}(\mathbf{x}, t | \mathbf{x}_0) p(\mathbf{x}_0 | \mathbf{x}_t = \mathbf{x}) d\mathbf{x}_0 = \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t=\mathbf{x}}[\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)]. \quad (19)$$

□

C Proofs of Main Results

C.1 Proof of Theorem 1 (Jacobian Variance Amplification)

Proof. Under the stop-gradient operator, the compound prediction V_{θ}^{sg} depends on θ only through $\mathbf{u}_{\theta}(\mathbf{x}_t, r, t)$. Let $\nabla_{\theta}\mathbf{u}_{\theta} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\theta}\ell_{\text{MF}} = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top}(\mathbf{r}_{\theta}^{\text{sg}} + \mathbf{J}\mathbf{v}'), \quad (20)$$

where $\mathbf{r}_{\theta}^{\text{sg}}$ is the mean-field residual and $\mathbf{J} = (t - r)\partial_{\mathbf{z}}\mathbf{u}_{\theta} - \mathbf{I}_d$ is the Jacobi factor. All three quantities— $\nabla_{\theta}\mathbf{u}_{\theta}$, $\mathbf{r}_{\theta}^{\text{sg}}$, and $\mathbf{J}$ —are deterministic conditioned on $\mathbf{x}_t$.

Since $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ by equation 5, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\theta}\ell | \mathbf{x}_t] = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{r}_{\theta}^{\text{sg}}. \quad (21)$$

The centered gradient is $\nabla_{\theta}\ell - \mathbb{E}[\nabla_{\theta}\ell \mid \mathbf{x}_t] = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{J}\mathbf{v}'$, so the conditional variance is:

$$\begin{aligned} \text{Var}[\nabla_{\theta}\ell \mid \mathbf{x}_t] &= \mathbb{E}\left[\|2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{J}\mathbf{v}'\|^2 \mid \mathbf{x}_t\right] \\ &= 4 \mathbb{E}\left[\left(\mathbf{v}'\right)^{\top} \mathbf{J}^{\top} \underbrace{(\nabla_{\theta}\mathbf{u}_{\theta})(\nabla_{\theta}\mathbf{u}_{\theta})^{\top}}_{\mathbf{G}_{\theta} \succeq \mathbf{0}} \mathbf{J}\mathbf{v}' \mid \mathbf{x}_t\right] \\ &= 4 \text{Tr}(\mathbf{G}_{\theta} \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}) \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}), \end{aligned} \quad (22)$$

where $\mathbf{G}_{\theta} = (\nabla_{\theta}\mathbf{u}_{\theta})(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses $\mathbf{G}_{\theta} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. $\square$

C.2 Proof of Semi-Gradient Gap

Proof. The MeanFlow loss conditioned on $\mathbf{x}_t$ decomposes as (cf. equation 10):

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\ell_{\text{MF}}] = \|\mathbf{r}_{\theta}^{\text{sg}}\|^2 + \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (23)$$

Without stop-gradient. Both terms depend on θ (through $\mathbf{u}_{\theta}$ and $\partial_z \mathbf{u}_{\theta}$), so the full gradient is:

$$\nabla_{\theta} \mathbb{E}[\ell] = \nabla_{\theta} \|\mathbf{r}_{\theta}\|^2 + \nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (24)$$

With stop-gradient. The JVP terms in $\mathbf{r}_{\theta}^{\text{sg}}$ and $\mathbf{J}$ are treated as constants, so:

$$\nabla_{\theta}^{\text{sg}} \|\mathbf{r}_{\theta}^{\text{sg}}\|^2 = 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{r}^{\text{sg}}, \quad \nabla_{\theta}^{\text{sg}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}) = \mathbf{0}. \quad (25)$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} \nabla_{\theta} \mathbb{E}[\ell] - \nabla_{\theta}^{\text{sg}} \mathbb{E}[\ell] &= \underbrace{\nabla_{\theta} \|\mathbf{r}_{\theta}\|^2 - 2(\nabla_{\theta}\mathbf{u}_{\theta})^{\top} \mathbf{r}^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})}_{\text{variance-driven difference}}. \end{aligned} \quad (26)$$

Expanding the first term: $\nabla_{\theta} \|\mathbf{r}\|^2 = 2(\nabla_{\theta}\mathbf{r}_{\theta})^{\top} \mathbf{r}_{\theta}$, and $\nabla_{\theta}\mathbf{r}_{\theta}$ includes the derivative of the JVP terms $(t-r)(\partial_z \mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\theta})$ w.r.t. θ , yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\theta}(\partial_z \mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\theta}) \cdot \mathbf{r}_{\theta}]$. At convergence $\mathbf{r}_{\theta} \rightarrow \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})$. $\square$

C.3 Proof of Theorem 3 (Flow Acceleration Expansion)

Proof. Let $\mathbf{x}_{\tau}$ solve the ODE $\frac{d\mathbf{x}_{\tau}}{d\tau} = \mathbf{v}(\mathbf{x}_{\tau}, \tau)$ with terminal condition $\mathbf{x}_t = \mathbf{x}_t$. The curvature gap $\Delta = \mathbf{u}(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t)$ can be written as:

$$\Delta(\mathbf{x}_t, r, t) = \frac{1}{t-r} \int_r^t \mathbf{v}(\mathbf{x}_{\tau}, \tau) d\tau - \mathbf{v}(\mathbf{x}_t, t) = \frac{1}{t-r} \int_r^t [\mathbf{v}(\mathbf{x}_{\tau}, \tau) - \mathbf{v}(\mathbf{x}_t, t)] d\tau. \quad (27)$$

Define $h = \tau - t \in [-(t-r), 0]$. Since $\mathbf{v} \in C^2$, we Taylor-expand $\mathbf{v}(\mathbf{x}_{\tau}, \tau)$ about $(\mathbf{x}_t, t)$ along the ODE trajectory:

$$\mathbf{v}(\mathbf{x}_{\tau}, \tau) = \mathbf{v}(\mathbf{x}_t, t) + \frac{D\mathbf{v}}{Dt}(\mathbf{x}_t, t) h + \frac{1}{2} \frac{D^2\mathbf{v}}{Dt^2}(\mathbf{x}_t, t) h^2 + O(h^3), \quad (28)$$

where $\frac{D}{Dt}$ denotes the material derivative along the flow. Substituting and integrating over $h \in [-(t-r), 0]$:

$$\begin{aligned} \Delta &= \frac{1}{t-r} \int_{-(t-r)}^0 \left[\frac{D\mathbf{v}}{Dt} h + \frac{1}{2} \frac{D^2\mathbf{v}}{Dt^2} h^2 + O(h^3) \right] dh \\ &= \frac{1}{t-r} \left[\frac{D\mathbf{v}}{Dt} \cdot \frac{h^2}{2} \Big|_{-(t-r)}^0 + \frac{1}{2} \frac{D^2\mathbf{v}}{Dt^2} \cdot \frac{h^3}{3} \Big|_{-(t-r)}^0 + O((t-r)^4) \right] \\ &= \frac{1}{t-r} \left[-\frac{(t-r)^2}{2} \frac{D\mathbf{v}}{Dt} + \frac{(t-r)^3}{6} \frac{D^2\mathbf{v}}{Dt^2} + O((t-r)^4) \right] \\ &= -\frac{(t-r)}{2} \frac{D\mathbf{v}}{Dt}(\mathbf{x}_t, t) + \frac{(t-r)^2}{6} \frac{D^2\mathbf{v}}{Dt^2}(\mathbf{x}_t, t) + \mathcal{R}_3(\mathbf{x}_t, r, t), \end{aligned} \quad (29)$$

where $\mathcal{R}_3 = O((t-r)^3)$ is the remainder. $\square$

C.4 Proof of ?? (Lipschitz Bound on the Curvature Gap)

Proof. From theorem 3, to leading order $\|\Delta\| \leq \frac{(t-r)}{2} \|\frac{Dv}{Dt}\| + O((t-r)^2)$. The material derivative satisfies:

$$\left\| \frac{Dv}{Dt} \right\| = \|\partial_t v + (\nabla_x v) v\| \leq \|\partial_t v\| + \|\nabla_x v\| \|v\| \leq L_t + L_x \|v(x_t, t)\|, \quad (30)$$

where the first inequality is the triangle inequality and the second follows from the Lipschitz assumptions $\|\partial_t v\| \leq L_t$ and $\|\nabla_x v\| \leq L_x$. $\square$

C.5 Proof of Proposition 5 (Gradient of VA-MF Loss)

Proof. The EMA velocity anchor $u_{\bar{\theta}}(x_t, t, t)$ is deterministic given x_t , since the EMA parameters θ are fixed during the gradient step. Under stop-gradient, V_{θ}^{EMA} depends on θ only through $u_{\theta}(x_t, r, t)$. By the same argument as equation 20, the per-step gradient is:

$$\nabla_{\theta} \ell = 2 (\nabla_{\theta} u_{\theta})^{\top} (V_{\theta}^{\text{EMA}} - v_{\text{cond}}). \quad (31)$$

Writing $v_{\text{cond}} = v(x_t, t) + v'$ and noting that $\tilde{r}^{\text{EMA}} := V_{\theta}^{\text{EMA}} - v(x_t, t)$ is deterministic given x_t :

$$\mathbb{E}_{x_0|x_t}[\nabla_{\theta} \ell] = 2 (\nabla_{\theta} u_{\theta})^{\top} \tilde{r}^{\text{EMA}}, \quad (32)$$

since $\mathbb{E}[v' | x_t] = 0$. The tower property $\nabla_{\theta} \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r,t,x_t}[\mathbb{E}_{x_0|x_t}[\nabla_{\theta} \ell]]$ gives the stated result. Crucially, no Jv' term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 20 where the stochastic tangent produces the Jv' noise. $\square$

C.6 Proof of Proposition 6 (Bias-Variance Tradeoff)

Proof. Expanding V_{θ}^{EMA} from equation 14, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\theta}^{\text{EMA}} = u_{\theta} + (t-r)(\partial_z u_{\theta} \cdot u_{\bar{\theta}}(x_t, t, t) + \partial_t u_{\theta}). \quad (33)$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$r_{\theta} = u_{\theta} + (t-r)(\partial_z u_{\theta} \cdot v(x_t, t) + \partial_t u_{\theta}) - v(x_t, t). \quad (34)$$

Subtracting:

$$\begin{aligned} \tilde{r}^{\text{EMA}} &= V_{\theta}^{\text{EMA}} - v(x_t, t) \\ &= r_{\theta} + (t-r) \partial_z u_{\theta} (u_{\bar{\theta}}(x_t, t, t) - v(x_t, t)). \end{aligned} \quad (35)$$

At convergence, the true residual $r_{\theta} \rightarrow 0$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 15 supervises the boundary condition, driving $u_{\bar{\theta}}(x_t, t, t) \rightarrow v(x_t, t)$ and hence $\tilde{r}^{\text{EMA}} \rightarrow 0$. $\square$

C.7 Proof of Proposition 7 (Variance Head Properties)

Proof. (a) **Gradient equivalence.** Under stop-gradient on the JVP, V_{θ}^{EMA} depends on θ only through u_{θ} . Differentiating the NLL equation 17 w.r.t. the mean parameters:

$$\nabla_{\theta}^{\text{mean}} \mathcal{L}_{\text{MF}}^{\text{NLL}} = \mathbb{E} \left[\frac{(\nabla_{\theta} u_{\theta})^{\top} (V_{\theta}^{\text{EMA}} - v_{\text{cond}})}{\sigma_{\theta}^2} \right]. \quad (36)$$

Since σ_{θ}^2 is deterministic given (x_t, r, t) and $\mathbb{E}_{x_0|x_t}[v'] = 0$, the v' cross term vanishes upon conditioning, yielding $\mathbb{E}[(\nabla_{\theta} u_{\theta})^{\top} \tilde{r}^{\text{EMA}} / \sigma_{\theta}^2]$.

(b) **Optimal variance.** Setting $\frac{\partial}{\partial \sigma^2} \mathcal{L}_{\text{MF}}^{\text{NLL}} = 0$ pointwise:

$$-\frac{\|V_{\theta}^{\text{EMA}} - v_{\text{cond}}\|^2}{2(\sigma^2)^2} + \frac{d}{2\sigma^2} = 0 \implies \sigma^{2*} = \frac{\|V_{\theta}^{\text{EMA}} - v_{\text{cond}}\|^2}{d}. \quad (37)$$

Taking $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}$ and writing $V_{\theta}^{\text{EMA}} - \mathbf{v}_{\text{cond}} = \tilde{\mathbf{r}}^{\text{EMA}} - \mathbf{v}'$ with $\mathbb{E}[\mathbf{v}'] = \mathbf{0}$:

$$\sigma^{2*} = \frac{\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\|\tilde{\mathbf{r}}^{\text{EMA}} - \mathbf{v}'\|^2]}{d} = \frac{\|\tilde{\mathbf{r}}^{\text{EMA}}\|^2 + \text{Tr}(\Sigma_{\mathbf{v}'})}{d}, \quad (38)$$

where the cross term $\tilde{\mathbf{r}}^{\text{EMA}\top} \mathbb{E}[\mathbf{v}'] = 0$ vanishes. At convergence $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$, giving $\sigma^{2*} \rightarrow \text{Tr}(\Sigma_{\mathbf{v}'})/d$.

(c) Automatic curriculum. By proposition 6, $\|\tilde{\mathbf{r}}^{\text{EMA}}\| \geq \|\mathbf{r}_{\theta}\|$ and by theorem 3, $\|\mathbf{r}_{\theta}\| \approx \|\Delta\| \propto (t-r)$ to leading order. Therefore σ^{2*} is increasing in $(t-r)$, and the effective weight $1/\sigma_{\theta}^2$ is smaller for large intervals—an automatic progressive curriculum. $\square$

D Additional Results

D.1 Acceleration-to-Noise Ratio from the Variance Head

The variance head from proposition 7 implicitly learns the flow acceleration structure identified in theorem 3.

Proposition 1 (ANR from the Variance Head). *Define the excess variance ratio $\rho(\mathbf{x}_t, r, t) := \sigma_{\theta}^2(\mathbf{x}_t, r, t)/\sigma_{\theta}^2(\mathbf{x}_t, t, t) - 1$. Near convergence:*

$$\rho \approx \frac{\|\Delta(\mathbf{x}_t, r, t)\|^2}{\text{Tr}(\Sigma_{\mathbf{v}'})} \approx \frac{(t-r)^2}{4} \cdot \frac{\|\frac{D\mathbf{v}}{Dt}(\mathbf{x}_t, t)\|^2}{\text{Tr}(\Sigma_{\mathbf{v}'})} = \frac{\text{ANR}(\mathbf{x}_t, t, r)^2}{4}, \quad (39)$$

where $\text{ANR} := (t-r) \|\frac{D\mathbf{v}}{Dt}\| / \sqrt{\text{Tr}(\Sigma_{\mathbf{v}'})}$ is the acceleration-to-noise ratio.

Proof. The FM anchor equation 15 ensures $\tilde{\mathbf{r}}^{\text{EMA}}(\mathbf{x}_t, t, t) \approx \mathbf{0}$ near convergence (the boundary condition is well-supervised). By proposition 7(b), $\sigma^{2*}(\mathbf{x}_t, t, t) \approx \text{Tr}(\Sigma_{\mathbf{v}'})/d$. For $r < t$, $\tilde{\mathbf{r}}^{\text{EMA}} \approx \Delta(\mathbf{x}_t, r, t)$, so $\sigma^{2*}(\mathbf{x}_t, r, t) \approx (\|\Delta\|^2 + \text{Tr}(\Sigma_{\mathbf{v}'}))/d$. Their ratio gives:

$$1 + \rho = \frac{\sigma^{2*}(\mathbf{x}_t, r, t)}{\sigma^{2*}(\mathbf{x}_t, t, t)} \approx \frac{\|\Delta\|^2 + \text{Tr}(\Sigma_{\mathbf{v}'})}{\text{Tr}(\Sigma_{\mathbf{v}'})} \implies \rho \approx \frac{\|\Delta\|^2}{\text{Tr}(\Sigma_{\mathbf{v}'})}. \quad (40)$$

Substituting $\|\Delta\| \approx \frac{(t-r)}{2} \|\frac{D\mathbf{v}}{Dt}\|$ from theorem 3 yields equation 39. $\square$

Proposition 1 shows that the excess variance ratio ρ directly measures the squared acceleration-to-noise ratio, connecting the learned variance head to the curvature-variance principle. This provides a testable prediction: for a trained model, $\rho(\mathbf{x}_t, r, t)$ should scale quadratically with $(t-r)$, with a slope proportional to $\|\frac{D\mathbf{v}}{Dt}\|^2 / \text{Tr}(\Sigma_{\mathbf{v}'})$.

D.2 Compound Prediction Error: d/dt vs. d/ds Formulations

The design space of two-parameter flow training objectives has a fundamental fork: which time variable to differentiate. Here we formalize the tradeoff between the d/dt formulation (used in MeanFlow and this work) and the d/ds formulation (used in TVM [48]).

Proposition 2 (Compound Prediction Error Comparison). *Consider two compound predictions under stop-gradient:*

(d/dt, **ours**): $V_{\theta}^{dt} = \mathbf{u}_{\theta}(\mathbf{x}_t, r, t) + (t-r) \text{sg}[JVP(\mathbf{u}_{\theta}, (\mathbf{x}_t, r, t), (\hat{\mathbf{v}}_t, 0, 1))]$, where $\hat{\mathbf{v}}_t$ is an estimate of $\mathbf{v}(\mathbf{x}_t, t)$.

(d/ds, **TVM [48]**): $V_{\theta}^{ds} = F_{\theta}(\mathbf{x}_t, t, s) + (s-t) \text{sg}[\partial_s F_{\theta}(\mathbf{x}_t, t, s)]$.

The mean-field residuals satisfy:

$$\|\tilde{\mathbf{r}}^{dt}\| \leq \|\mathbf{r}_{\text{true}}\| + (t-r) \|\partial_{\mathbf{z}} \mathbf{u}_{\theta}\| \|\hat{\mathbf{v}}_t - \mathbf{v}(\mathbf{x}_t, t)\|, \quad \|\tilde{\mathbf{r}}^{ds}\| \leq \|\mathbf{r}_{\text{true}}^{ds}\|. \quad (41)$$

The d/ds formulation has no tangent estimation error because $\partial_s F_{\theta}$ is a partial derivative w.r.t. a network input—no spatial Jacobian or velocity estimate is needed.

Proof. For d/dt : by proposition 6, $\tilde{\mathbf{r}}^{dt} = \mathbf{r}_{\text{true}} + (t-r) \partial_{\mathbf{z}} \mathbf{u}_{\theta}(\hat{\mathbf{v}}_t - \mathbf{v})$. The triangle inequality gives the upper bound. The tangent estimation error $(t-r) \|\partial_{\mathbf{z}} \mathbf{u}_{\theta}\| \|\hat{\mathbf{v}}_t - \mathbf{v}\|$ shrinks as the EMA model improves ($\hat{\mathbf{v}}_t \rightarrow \mathbf{v}$).

For d/ds : the JVP tangent is $\partial_s F_{\theta}$, a partial derivative of the network w.r.t. its third scalar input s . This is deterministic and exact—no velocity field estimate is needed. The compound prediction error reduces to the true identity residual $\mathbf{r}_{\text{true}}^{ds}$. $\square$

Tradeoff. The d/ds formulation eliminates the spatial Jacobian entirely (a strict advantage in compound prediction error), but requires parameterizing $f_{\theta}(\mathbf{x}_t, t, s) = (s-t)F_{\theta}(\mathbf{x}_t, t, s)$ with explicit dependence on both time endpoints—a different architecture from standard velocity networks. The d/dt formulation retains compatibility with the standard MeanFlow parameterization and benefits from the curvature-variance theory developed in this work. The bias term $(t-r) \|\partial_{\mathbf{z}} \mathbf{u}_{\theta}\| \|\hat{\mathbf{v}}_t - \mathbf{v}\|$ shrinks during training, so the gap between the two formulations closes as the EMA anchor improves.

D.3 MSE Variant without Variance Head

As a simpler alternative to Algorithm 1, one can use explicit weighting $w(t) \geq 0$ in place of the heteroscedastic loss. A principled choice is the signal-to-noise ratio schedule $w(t) = (1-t)^2/(t^2 + \epsilon)$, which captures the qualitative variance structure for linear interpolation paths by downweighting high-variance timesteps near $t = 1$.

Algorithm 2 VA-MeanFlow Training (MSE Variant)

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$, weighting $w(t)$
Initialize $\bar{\theta} \leftarrow \theta$
for $k = 1, 2, \dots$ **do**
 Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
 $\mathbf{x}_t \leftarrow (1-t)\mathbf{x}_0 + t\mathbf{x}_1$
 $\mathbf{v}_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)]$ ▷ EMA anchor
 $V_{\theta} \leftarrow \mathbf{u}_{\theta}(\mathbf{x}_t, r, t) + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (\mathbf{x}_t, r, t), (\mathbf{v}_{\text{tang}}, 0, 1))]$
 $\ell_{\text{MF}} \leftarrow w(t) \|\mathbf{V}_{\theta} - \mathbf{v}_{\text{cond}}\|_2^2$
 $\ell_{\text{FM}} \leftarrow \|\mathbf{u}_{\theta}(\mathbf{x}_t, t-\delta, t) - \mathbf{v}_{\text{cond}}\|_2^2$ ▷ FM anchor
 $\theta \leftarrow \theta - \eta \nabla_{\theta} (\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$
 $\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$
end for
return θ ▷ Generate: $\hat{\mathbf{x}}_0 = \mathbf{x}_1 - \mathbf{u}_{\theta}(\mathbf{x}_1, 0, 1)$
